

Department of Computer Science and Engineering
Academic Year 2020– 2021 (Odd Semester)

Degree, Semester & Branch: B.E, V, CSE

Course Code & Title: CS501 & Theory of Computation

Name of the Faculty member (s): Dr.I.Gethzi Ahila Poornima

Innovative Practice Description

- **Unit / Topic: Unit II / Closure properties of Regular Languages**
- **Course Outcome: CO2**
- **Topic Learning Outcome: TLO3**
- **Activity Chosen: One Minute Paper**
- **Justification:**

The closure properties of regular language are more important to prove various theorems in automata. By carrying out this practice, the student's understanding level is identified and taught to them again.

- **Time Allotted for the Activity: 25 Minutes**
- **About the Event:**

A particular topic whose understanding is very important in the point of examinations and further understanding of the upcoming topics is chosen. This topic will be taught in the first 25 minutes of the class. At the end of the class, to analyse the level of understanding of the students, the students will be asked to write a few points regarding the topic taught in the class. From their write-up, the course instructor could know the level of understanding of students.

- **Details of the Implementation:**

The closure properties of regular languages were taught in the first 25 minutes. At the end of the class, the students (54) were asked to write a few closure properties of regular languages. The students wrote the closure properties up to their understanding level. The content that they had written was verified at the end. After evaluating the one-minute paper written, the understanding level of the students were identified. Around 30 % of students haven't written homomorphism and inverse homomorphism property. The same was explained again in the next class. A sample page written by Ms.B. Dharini is given in Figure 1.

- **CO – PO / PSO mapping:**

CO	PO2	PO3	PSO1
CO2	1	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO2	PO3	PSO1
	1	2	1
Justification for correlation	Students could formulate complex engineering problems by expressing them in a standard notation such as regular expression and automata which models the system.	Students will be able to describe the language using regular expression which is useful for several kinds of software including compiler.	By formulating regular expressions and FA, students will be able to develop various software components especially system software components

• **Images / Screenshot of the practice:**

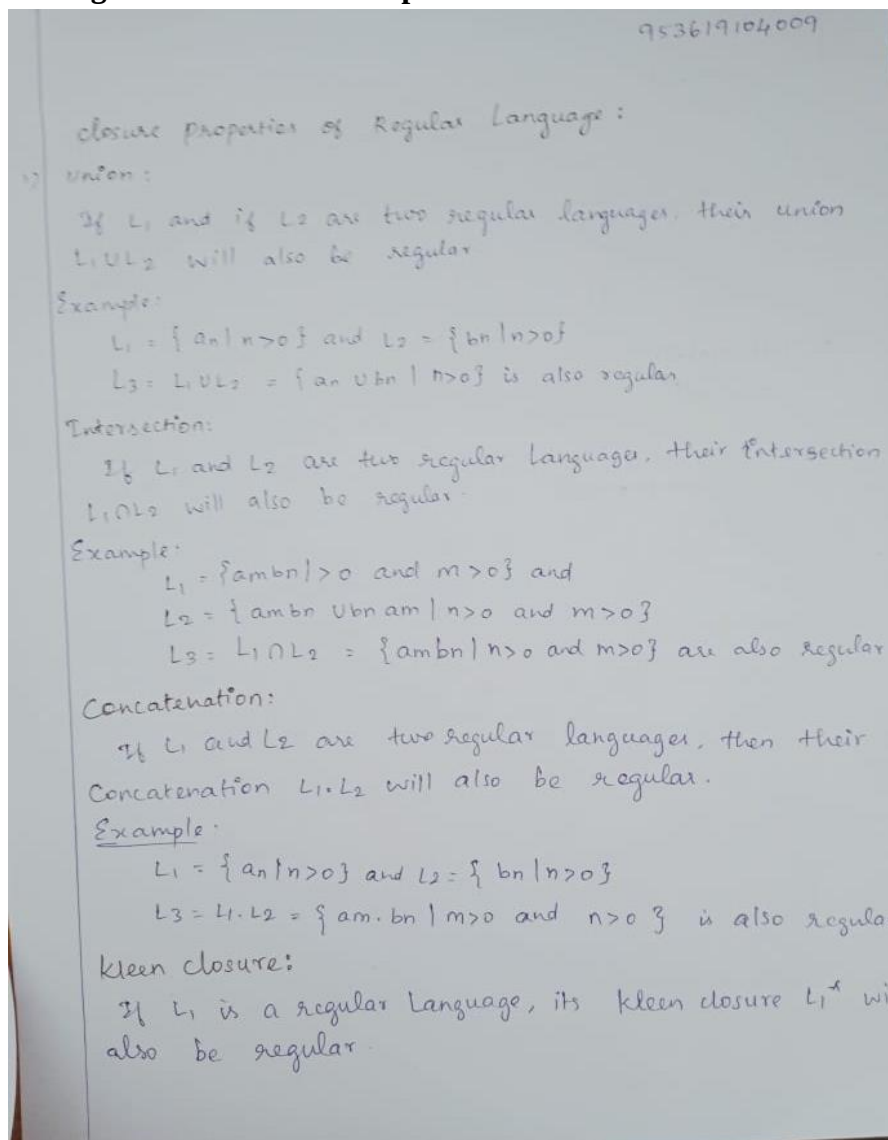


Figure 1- Sample Page written by Ms.Dharini (953619104009)

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- ❖ The students gave a very good feedback about this activity. They felt that, they could be more thorough with this concept by doing this activity.

- ❖ ***Benefit of the practice:***

- Most of the students take this activity as positive and started writing about the concept in the paper.
 - It can be easily tracked whether the students have understood the concept clearly or not.
 - Since this concept is the most important one for this course, this activity will make us to identify the understanding level of the students on this particular concept.

- ❖ ***Challenges faced in implementation:***

- Very few students are not confident in their understanding towards this concept. They hesitated to prepare a write up on this concept.

Signature of Faculty Member

HOD